

# **ER DIAGRAM DESIGN AND RELATIONAL SCHEMA**

## **Assignment 2 : Hotel Management System**

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## OBJECTIVE

To design a complete Entity Relationship (ER) Diagram for the Hotel Management System and convert it into a Relational Schema by identifying entities, attributes, primary keys, foreign keys, relationships, and cardinalities.

## ENTITIES AND ATTRIBUTES

### Entity 1: GUEST

| Column     | Key         |
|------------|-------------|
| guest_id   | Primary Key |
| guest_name | -           |
| email      | -           |
| phone      | -           |
| address    | -           |

### Entity 2: ROOM\_TYPE

| Column          | Key         |
|-----------------|-------------|
| room_type_id    | Primary Key |
| type_name       | -           |
| price_per_night | -           |
| description     | -           |

### Entity 3: ROOM

| Column       | Key         |
|--------------|-------------|
| room_id      | Primary Key |
| room_number  | -           |
| floor_number | -           |
| room_status  | -           |
| room_type_id | Foreign Key |

### Entity 4: STAFF

| Column     | Key         |
|------------|-------------|
| staff_id   | Primary Key |
| staff_name | -           |
| role       | -           |
| phone      | -           |
| department | -           |

### Entity 5: RESERVATION

| Column             | Key         |
|--------------------|-------------|
| reservation_id     | Primary Key |
| guest_id           | Foreign Key |
| room_id            | Foreign Key |
| check_in_date      | -           |
| check_out_date     | -           |
| reservation_status | -           |

### Entity 6: SERVICE

| Column        | Key         |
|---------------|-------------|
| service_id    | Primary Key |
| service_name  | -           |
| service_price | -           |
| description   | -           |

### Entity 7: PAYMENT

| Column         | Key         |
|----------------|-------------|
| payment_id     | Primary Key |
| reservation_id | Foreign Key |
| payment_date   | -           |
| amount         | -           |
| payment_method | -           |
| payment_status | -           |

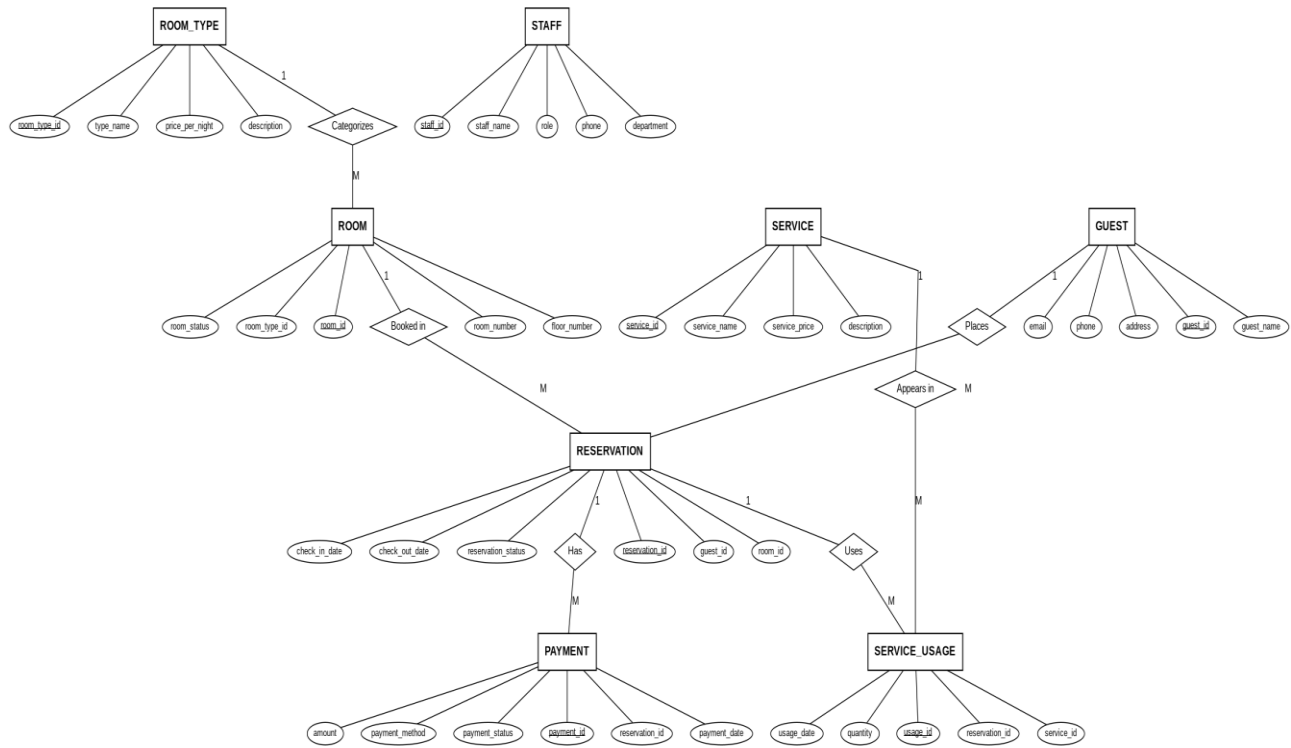
### Entity 8: SERVICE\_USAGE

| Column         | Key         |
|----------------|-------------|
| usage_id       | Primary Key |
| reservation_id | Foreign Key |
| service_id     | Foreign Key |
| usage_date     | -           |
| quantity       | -           |

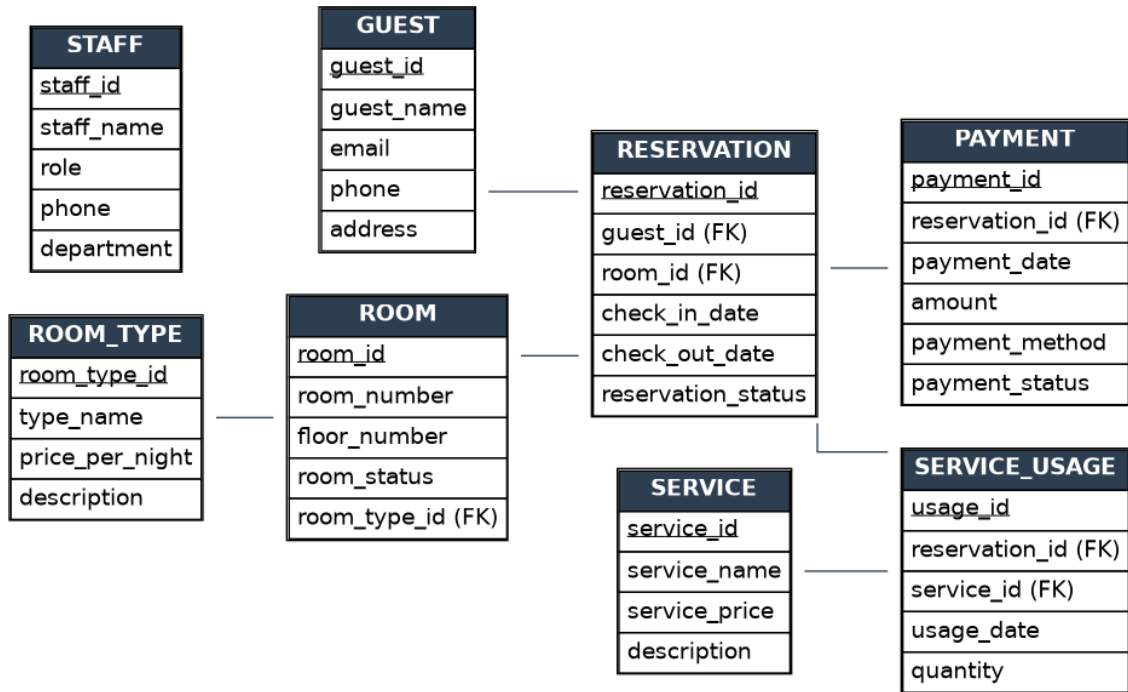
## RELATIONSHIPS AND CARDINALITIES

| Entity 1    | Relationship | Entity 2      | Cardinality |
|-------------|--------------|---------------|-------------|
| GUEST       | Places       | RESERVATION   | 1:M         |
| ROOM_TYPE   | Categorizes  | ROOM          | 1:M         |
| ROOM        | Booked in    | RESERVATION   | 1:M         |
| RESERVATION | Has          | PAYMENT       | 1:M         |
| RESERVATION | Uses         | SERVICE_USAGE | 1:M         |
| SERVICE     | Appears in   | SERVICE_USAGE | 1:M         |

## ER DIAGRAM



## RELATIONAL SCHEMA



## PRIMARY KEYS AND FOREIGN KEYS

| Table         | Primary Key    | Foreign Key                |
|---------------|----------------|----------------------------|
| GUEST         | guest_id       | -                          |
| ROOM_TYPE     | room_type_id   | -                          |
| ROOM          | room_id        | room_type_id               |
| STAFF         | staff_id       | -                          |
| RESERVATION   | reservation_id | guest_id, room_id          |
| SERVICE       | service_id     | -                          |
| PAYMENT       | payment_id     | reservation_id             |
| SERVICE_USAGE | usage_id       | reservation_id, service_id |

## RELATIONSHIP EXPLANATION

**Guest – Reservation:** One guest can place many reservations. Therefore, the relationship is 1:M.

**Room Type – Room:** One room type can be assigned to many rooms. Therefore, the relationship is 1:M.

**Room – Reservation:** One room can be booked in many reservations over time. Therefore, the relationship is 1:M.

**Reservation – Payment:** One reservation can have one or more payment records in this model. Therefore, the relationship is 1:M.

**Reservation – Service\_Usage:** One reservation can include many service-usage records. Therefore, the relationship is 1:M.

**Service – Service\_Usage:** One service can appear in many service-usage records. Therefore, the relationship is 1:M.